Project Summary

Title: *Clinical ML & Explainable AI for 30-Day Diabetes Readmission Risk Prediction*

The project uses the **UCI Diabetes 130-US Hospitals dataset** to develop machine-learning models that predict whether a patient will be **readmitted within 30 days**.

The notebook follows an end-to-end clinical ML workflow:

- Loads and inspects the diabetes hospital dataset.
- Handles missing values, including converting ? to missing values.
- Removes duplicate patients and excludes certain expired/hospice discharge cases.
- Removes high-missingness variables such as weight, payer_code, medical_specialty, and max_glu_serum.
- Defines the target as **30-day readmission (<30) vs. not 30-day readmission**.
- Removes identifiers and the original readmission variable to reduce leakage.
- Performs exploratory analysis of hospital stay, previous inpatient visits, medication count, and age.
- Creates engineered features including **total prior healthcare utilization, ordinal age, and service density**.
- Uses an **80/20 stratified train-test split**.
- Applies median imputation and scaling to numerical variables and categorical imputation/one-hot encoding to categorical variables.
- Trains and compares **Logistic Regression, Random Forest, and XGBoost** models.
- Evaluates models using **ROC-AUC, PR-AUC, precision, recall, F1-score, specificity, and Brier score**.
- Performs ROC and Precision-Recall analysis.
- Examines different prediction thresholds from **0.10 to 0.90**.
- Evaluates probability calibration.
- Uses **SHAP** to explain XGBoost predictions through global and individual-patient explanations.

- Performs subgroup analysis by **gender and age group**.
- Conducts false-positive and false-negative error analysis.
- Produces a final automated summary of the model results and important predictive factors.

Main research focus

The project ultimately investigates whether **clinical and healthcare-utilization information can be used to identify patients at higher risk of 30-day hospital readmission**, while also addressing **explainability, calibration, class imbalance, model performance, and potential clinical limitations**.

One important point: the notebook's final summary identifies **prior inpatient utilization, length of stay, medication burden, and service density** as major predictive factors. These should be described as **model-associated predictive factors, not causal clinical risk factors**.